# CAGE-Fake: Article Trustworthiness Audit Report
## Calibration-Aware Gating & Evidence-Grounded Reasoning for Trustworthy Fake News Triage

**AUDIT ID:** AUD-205621
**DATE:** 20261028_220957
**FRAMEWORK:** CAGE-Fake v2.0

**LEGAL & FACT-CHECKING POLICY:** A trained fake-news classifier predicts linguistic, stylometric, and semantic patterns learned from benchmark datasets; it does **NOT** independently verify factual reality. URL extraction, pattern recognition, and external evidence verification remain strictly separate processes. The pipeline abstains (**UNCERTAIN**) whenever evidence is missing, conflicting, or model consensus is low.

**Title:** Man gets life term for abducting, raping minor
**Source:** https://timesofindia.indiatimes.com/city/agra/man-gets-life-te...
**Word Count:** 215 words | **Extraction:** Successfully Extracted via Trafilatura
**OOD Detector:** Loaded (SBERT & Vocab) | **Triage Policy:** Evidence-Grounded Three-Way Gating

**TRIAGE VERDICT: UNCERTAIN**
**Abstention Confidence:** 50.0% ($\tau$ = 80.0%)
**Model Disagreement (std):** 0.208 ($\delta$ = 0.15)
**Safety Trigger:** Stylometric vs Lexical Split (RF Dissent)

### 1. Multi-Model Disagreement Analysis (5 Models)

| Model Name | P(FAKE) | Vote | Confidence |
|---|---|---|---|
| Random Forest | 42.1% | REAL | 57.9% |
| XGBoost | 93.9% | FAKE | 93.9% |
| LightGBM | 91.1% | FAKE | 91.1% |
| Logistic Regression | 69.5% | FAKE | 69.5% |
| CAGE-Fake Stack | 93.8% | FAKE | 93.8% |

**Consensus Status:** High Dissent / Conflicting Predictions
**Vote Distribution:** **4 FAKE** vs **1 REAL**
**Base Disagreement (4-base std):** 0.208 ($\delta$ = 0.15)
**5-Model Consensus Spread:** 0.202
[==========------]
**Safety Assessment:** Base-model disagreement (0.208) exceeds safety threshold delta (0.150).

### 2. News Credibility Fingerprint (Linguistic & Stylistic Scorecard)

| Credibility Dimension | Score (0-100) | Journalistic Assessment |
|---|---|---|
| Sensationalism Discipline | 90.3/100 | Excellent / Restrained |
| Source Attribution | 75.0/100 | Adequate / Neutral |
| Epistemic Grounding | 90.3/100 | Excellent / Restrained |
| Punctuation Discipline | 88.2/100 | Excellent / Restrained |
| Readability Structure | 90.3/100 | Excellent / Restrained |
| Lexical Diversity | 79.0/100 | Adequate / Neutral |

OVERALL CREDIBILITY INDEX
**85.1 / 100**
**Credibility Tier:** A (Restrained / Professional Copy)

Scores are standardized against verified newsroom copy standards: high factual attribution, balanced syntax, and low sensationalism maximize the index.

### 3. Grounded Diagnostic Reasoning & Counterfactual Sensitivity

**Executive Synthesis:** System abstains with UNCERTAIN triage (Base models split on verdict with stylometric RF expert dissenting (RF P(FAKE)=42.1%, but LR=69.5%, LGBM=91.1%, XGB=93.9%). Lexical and semantic models key on out-of-domain vocabulary while stylometric expert dissents.). Multi-view divergence (std=0.208) mandates human editorial review.

**Diagnostic Explanation:** 1. Decision & Triage Status: The CAGE-Fake pipeline reached a verdict of **UNCERTAIN** with **50.0% calibrated confidence**.
- **Gating Trigger**: System abstention triggered under safety policy (style_lexical_split).

**Counterfactual Simulation:** Neutralizing 1 linguistic markers resulted in a signed probability shift of -3.4 percentage points (from 93.8% to 90.3%), retaining the text classifier's verdict (FAKE) (note: the final triage verdict may differ when evidence overrides it). Original P(FAKE): **93.8%** → Transformed P(FAKE): **90.3%** **(Signed Shift: -3.4%)**

**[v] Clean Stylometrics:** No significant sensational exaggeration, clickbait patterns, or manipulative phrasing detected.

## 4. Claim-Level Semantic Evidence Grounding (Triage Verification)

| Extracted Factual Claim | Verification Status | Authoritative Source / Match | Evidence Context Quote |
|---|---|---|---|
| Agra: A special Pocso court in Agra has sentenced a 30-year-old man to life impris... | UNVERIFIED (Sim: 0.36) | No live sources >= 0.60 similarity | No corroborating external sources met the >= 0.60 semantic similarity floor. |
| During the trial, the prosecution examined 12 witnesses, including the complainant... | SUPPORTED (Sim: 0.78) | medicaldialogues.in | “During the trial, the court examined 12 prosecution witnesses, of whom six, inclu... |
| While delivering the verdict, special judge (Pocso) Sonika Chaudhary highlighted s... | SUPPORTED (Sim: 0.70) | hindustantimes.com | ![The special judge (POCSO Court), Sonika Chaudhary, observed that although young ... |
| Man gets life term for abducting, raping minor. | SUPPORTED (Sim: 0.60) | abc3340.com | # Life behind bars for man convicted of kidnapping and raping 10-year-old The Blo... |

## 5. Local Signal Attributions (Meta-Ensemble Impacts)

| Top FAKE-Driving Indicators | Impact | Top REAL-Driving Anchors | Impact |
|---|---|---|---|
| LR [TF-IDF Lexical] | +1.444 | RF [Stylometrics] | -0.223 |
| XGB [SBERT Semantic] | +0.855 | Inter-View Disagreement (std) | -0.208 |
| LightGBM [SVD Topics] | +0.574 | - | - |

## 6. Empirical Benchmark Model Comparison (Held-out Test Split)

| Model Name | Accuracy | Macro-F1 | ECE | Coverage | Selective Acc |
|---|---|---|---|---|---|
| Random Forest (Uncalibrated) | 0.8098 | 0.8097 | 0.0770 | 100.0% | 0.8098 |
| XGBoost (Uncalibrated) | 0.8508 | 0.8505 | 0.0550 | 100.0% | 0.8508 |
| LightGBM (Uncalibrated) | 0.8787 | 0.8785 | 0.0401 | 100.0% | 0.8787 |
| Logistic Regression (Uncalibrated) | 0.9361 | 0.9361 | 0.1322 | 100.0% | 0.9361 |
| Uncalibrated Stack | 0.9410 | 0.9410 | 0.0182 | 100.0% | 0.9410 |
| Calibrated Stack (Full Coverage) | 0.9410 | 0.9409 | 0.0274 | 100.0% | 0.9410 |
| **CAGE-Fake (Calibrated + Gated)** | **0.9410** | **0.9409** | **0.0274** | **82.5%** | **0.9781** |

## 7. Research Calibration & Disagreement Visualizations










